

Aerodynamic Design Optimization of Wing-Sails

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An eco-friendly, alternative energy source is a key topic in all engineering fields. In the ship industry, wing-sail technology is drawing great attention as a new type of propulsion that takes full advantage of free wind energy. The purpose of our study is to numerically analyze the aerodynamic characteristics of wing-sails and optimize their shape and operation conditions in terms of the angle of attack, flap length, and deflection angle under various wind directions. A viscous Navier-Stokes flow solver is used for the numerical aerodynamic analysis. A design optimization framework using an evolutionary algorithm and the Kriging surrogate model is developed and finds the optimum operation condition for multiple wing-sails to maximize the resultant propulsive force. First, the exact response surfaces of C_l and C_d are created for a single wing-sail, and a simple trade-off study is conducted to investigate the flap effects in terms of its deflection angle and flap length. Next, the proposed design optimization framework is applied to find the optimum shape and operating conditions for multiple wing-sails, which have three identical wing-sails in a row. A total of nine design variables are employed, including the angle of attack, flap length and deflection angle for each wing-sail, and relative wind direction, which is allowed to vary from 45° to 90° and 135° . Although the design is carried out using a two-dimensional topology, mostly because of the high computational cost related to the nature of gradient-free optimization algorithms, the design results are validated using a high-fidelity three-dimensional Computational Fluid Dynamics analysis. Predicted improvement in two-dimensional analysis of the averaged thrust by 14~22% of the baseline values turned out to be slightly lower and becomes 10~17% in full three-dimensional flow analysis. Although an initial set of the angle of attack values is varied from 8° to 9.5° to investigate the effects of the initial angle of attack, the trends of the design optimization are preserved. In summary, multiple wing-sails can enhance a ship's propulsion system by more than 10%.

Nomenclature

C_L	=	Lift coefficient
C_D	=	Drag coefficient
C_{fx}	=	Thrust coefficient along x-direction
$C_{fx,ave}$	=	Averaged thrust coefficient along x-direction
C_p	=	Surface pressure coefficient
α	=	AoA (Angle of attack)
δ	=	Flap angle
θ	=	Wind direction

I. Introduction

Energy is becoming a pressing issue in industry and academia because energy consumption has been growing continuously and is focused mainly on fossil fuels, which create environmental pollution. An eco-friendly energy source as an alternative to fossil fuels is actively sought in all engineering fields.

In the ship industry, a conventional propulsion system also greatly relies on fossil fuels and the emission of pollutants causes great problems in the ocean system. Therefore, new propulsion technologies have been developed

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that can replace existing reciprocating and turbine engines, and use eco-friendly alternative energy sources. One of these uses wind energy as an additional propulsive force. In fact, it has always been the main source of propulsion for sailing ships. The basic concept is to enhance the existing propulsion efficiently by using aerodynamic propulsion provided by sails or “wing-sails,” which utilize the same shape as the wing of an airplane rather than conventional sails made of fabric. Recent research results reached the conclusion that more than 50% of the existing propulsion force can be replaced with wing-sails, leading to a saving of an equivalent proportion of fossil fuels¹.

Research on wing-sails has been conducted by many groups. Kazuyuki Ouchi et al. performed aerodynamic and structure analyses of multiple wing-sails using numerical methods¹. Toshifumi Fujiwara et al. studied the aerodynamic characteristics of multiple wing-sails with respect to the angle of the boom and slot². Furthermore, Toshifumi Fujiwara et al. carried out a wind tunnel experiment to reveal the sail-sail and sail-hull interaction effects of a hybrid sail³. Takuji Nakashima et al. conducted a wind tunnel measurement and numerical simulation to clarify the aerodynamic interaction phenomena between wing-sails in a new conceptual wind-driven vessel⁴. However, most of these studies focused only on an aerodynamic analysis of the wing-sails and depended on wind tunnel experiments. Not much attention has been given to the design aspect of multiple wing-sails. In the present study, we conduct a high-fidelity based design optimization, which is tightly coupled with numerical analysis and optimization algorithms.

In this study, we perform a high-fidelity aerodynamic analysis using Computational Fluid Dynamics (CFD) flow solvers to investigate the aerodynamic characteristics of multiple wing-sails, and the highly accurate flow analysis is directly used for aerodynamic design optimization in consideration of flow interactions in the multiple wing-sails. A key advantage of our work is conduction of a trade-off study of the design parameters that are known to be effective in improving the wing-sail performance. We find the optimum topology of the multiple wing-sails in terms of the operation conditions and flap configuration. First, we conduct an aerodynamic analysis to investigate the effects of the flow interactions and aerodynamic characteristics of the multiple wing-sails. A viscous Navier-Stokes solver is used with an unstructured mesh topology for both two- and three-dimensional numerical analyses of the multiple wing-sails. The turbulence model of Spalart-Allmaras (SA) is employed. The magnitude of the propulsive force is investigated for the wing-sails both with and without the flap, while the wind direction is allowed to vary from 0° to 360° in 15° steps. To validate the accuracy of the CFD solver for the current flow conditions of the wing-sails, a two-dimensional flow analysis is conducted around the NACA0012 airfoil, and the results are compared with the experimental values. Next, an aerodynamic design optimization framework for multiple wing-sails is developed. This framework uses evolutionary algorithms and the Kriging surrogate model. The optimization framework is fully automated.

The evolutionary algorithm of the genetic algorithm (GA) is particularly attractive because it does not require gradient-based sensitivity information and is very robust and effective at finding the global optimum. Furthermore, the GA searches the entire design space and is less likely than a gradient-based method to get stuck at the local optimum. However, the GA requires a large number of function evaluations in each generation. To alleviate the computational cost of the optimization algorithm related to the large number of function evaluations, we employ the Kriging surrogate model, which is an interpolation-based surrogate model. A surrogate model, known as an approximate model or meta model, is an efficient way of alleviating an expensive computing burden. The Kriging surrogate model is created by evaluating an initial sampling set of function evaluations through high-fidelity Navier-Stokes flow solutions. We also carry out the high-fidelity CFD-based design optimization using a continuous adjoint method tightly coupled with the CFD solver and adjoint solver. Because we consider the operating condition as well as the flap configuration as design variables, it is difficult to smoothly handle the mesh deformation.

Using this design framework, we find the optimal setup for the multiple wing-sails to maximize the averaged thrust coefficient and thus increase the overall propulsive force. A total of nine design variables are employed, and each wing-sail has design parameters for varying the angle of attack of the wing-sail and the flap configurations in terms of the length and its deflection angles. In our previous study, we only considered the variation in the angle of attack of each wing-sail as design variables that could be rotated up to $\pm 15^\circ$ independently^{5,6}. However, in this study, we consider not only the angle of attack but also the flap length and deflection angle of each wing-sail as design variables. Finally, the optimal design variables are obtained to maximize the averaged thrust performance of the multiple wing-sails, and the wind direction is allowed to vary from 45° to 90° and 135°. We initially perform the design optimization at an angle of attack of 8° and later at 9.5°, because a 9.5° angle of attack is closer to a stall condition. As a result, we achieve a thrust improvement of approximately 14~22% at the initial angle of attack of AoA = 8° and 10~12% at AoA = 9.5°. Although the design is carried out based on the two-dimensional flow analysis, mostly because of the expensive computational cost, a validation is carried out with the complete three-dimensional wing-sails using a CFD analysis to determine whether the thrust improvement achieved from the two-dimensional assumption is valid. The results show that the thrust performance improvement in the multiple wing-

sails becomes slightly lower in the three-dimensional analysis (10~17% at $AoA = 8^\circ$ and 9~11% at $AoA = 9.5^\circ$), which is very predictable considering the three-dimensional aerodynamic effects, including cross-flows.

In conclusion, our work has advantages of providing information useful for understanding the aerodynamic characteristics of wing-sails and of determining the optimum setup for multiple wing-sails under various operation conditions. Our work indicates that the concept of wing-sails is a valid and viable tool that can serve as an efficient alternative propulsion system in the ship industry.

The outline of this paper is as follows. Section II introduces the concept of wing-sails, topology of multiple wing-sails, and force relation for evaluating the thrust performance. Sections III and IV respectively describe the aerodynamic analysis method and design optimization framework. Section IV discusses the Kriging approach for surrogate modeling, as well as the GA employed in this study. Section V explains the aerodynamic characteristics and flow interaction of multiple wing-sails. Section VI then shows the results of the aerodynamic design optimization, initially at an angle of attack of 8° and then at 9.5° in consideration of the flow interaction, as mentioned in Section V. Section VII concludes this paper.

II. Concept of Wing-Sails

A sailing ship is an eco-friendly and fuel-efficient vessel that uses wing-sails to support an existing fossil fuel-based propulsion system. The wing-sail technology uses the free wind energy available at sea to produce an additional thrust force. As shown in Fig. 1 (left), the “Shin-Aitoku Maru” was the first sailing ship to use a wing-sail propulsion system. It was introduced by JAMDA Japan in the 1970s. At first, simple flat-type sails were applied. Then, the sailing ship shown in the right-hand illustration of Fig. 1 was developed, which utilized sails with an airfoil shape instead of the flat-type of sail⁶. This considerably improved the aerodynamic performances of the wing-sails.



Fig. 1 Examples of wing-sail ships: Shin-Aitoku Maru (left)⁷ and UT Wind Challenger ship (right)⁸

The wing-sails are located in the limited space available on the main deck⁸. The number of wing-sails is generally determined by the size of the ship. In a previous study, a wing-sail system with a total of six sails was considered for a cargo ship in the 100-K DWT (Dead Weight Tons) class^{5,6}. In this study, we consider a wing-sail system with three sails for a cargo ship in the 50-K DWT class. The reduction in the number of wing-sails in the present study is based on the observation that the most dynamic flow interaction occurs in the area of the first three wing-sails. Fig. 2 shows a schematic of the wing-sail system being considered here. Each wing-sail has a chord length of 10 m, height of 20 m, and aspect ratio of 2 without any taper. In addition, the interval between the wing-sails is 15 m, which is 1.5 times the chord length. The three wing-sails are arranged along the x direction. The wing-sails can rotate to face various wind directions, and the axis of rotation is located at 40% of each chord length. The sectional shape of the wing-sails is an NACA 0012 airfoil, which is symmetric and does not have a camber. The reason for selecting the symmetric airfoil is to prevent any negative effects on the aerodynamic performance because the wind direction can vary from 0° to 360° during a cruise mission. To improve the aerodynamic performance of the wing-sails, as shown in Fig. 2, a flap system is considered as a high-lift device.

As shown in Fig. 2, if the wing-sails are constructed as multiple arrangements in a row to generate more thrust, a flow interaction occurs among them, and it has a significant effect on the aerodynamic performance of the multiple wing-sails. Therefore, in order to estimate the thrust of multiple wing-sails, the effect of this flow interaction should be considered. In our previous study^{5,6}, to investigate the effect of the flow interaction and the flow characteristics around multiple wing-sails, we performed a three dimensional aerodynamic analysis of six sails using a high-fidelity numerical analysis.

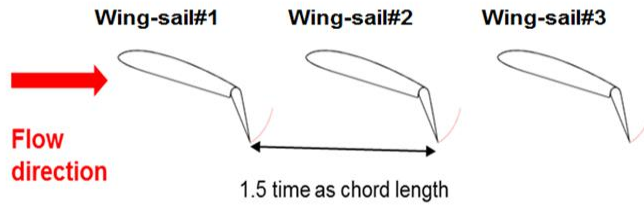


Fig. 2 Schematic view of multiple wing-sails

Fig. 3 shows a force diagram representing the sail direction, relative wind direction, and force coefficients acting on the wing-sail. Consider that the direction that the ship is being propelled in is parallel to the x direction. Then, the resultant force along the x-direction, which is thrust vector C_{fx} , can be calculated using Eq. (1). The formulation of the thrust vector can be defined as a function of the wind direction (θ) and aerodynamic coefficients (C_L, C_D) of the wing-sails. It is assumed that the resultant force along the y direction is canceled out by the rudders of a thrust vectoring system, such as azimuth thrusters, and we do not consider any side components of the force other than the propulsive force. The thrust generation principles are applied to each wing-sail. The thrust coefficients along both the x and y directions can be calculated using Eq. (1) and Eq. (2). For multiple wing-sails, the average thrust is considered to be a performance indicator, and it is an average thrust coefficient divided by the number of wing-sails. In this study, a wing-sail system with three sails is considered; thus, the average thrust is defined as Eq. (3).

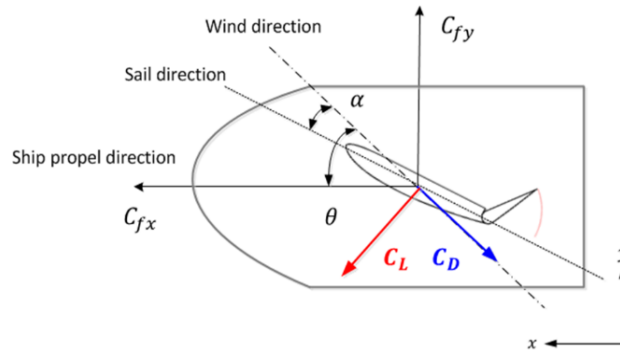


Fig. 3 Resultant force relation

$$C_{fx} = C_L \sin(\theta) - C_D \cos(\theta) \quad (1)$$

$$C_{fy} = C_L \cos(\theta) + C_D \sin(\theta) \quad (2)$$

$$C_{fx,ave} = \frac{1}{3} \sum_{i=1}^3 C_{fx,i} \quad (3)$$

III. Aerodynamic Analysis Methods

In order to investigate the flow phenomena, an appropriate CFD flow solver must be selected. SU²_CFD is fluid dynamics software for simulations in a parallel computation environment, along with a module for partitioning the volumetric grid as a pre-processor in parallel flow computations. SU²_CFD solver can provide direct flow solutions and adjoint solutions for potential, Euler, Navier-Stokes, and Reynolds Averaged Navier-Stokes (RANS) governing equations. It uses a Finite Volume Method (FVM) for spatial discretization. Both explicit and implicit methods are available for time integration, and central difference or upwind methods can also be used for spatial discretization. To improve the robustness and convergence of the flow solution, the advanced numerical techniques of residual smoothing and agglomeration multi-grid methods are also available⁹.

For the numerical analysis, we solve the two- and three-dimensional compressible RANS governing equations,

which describe the conservation of mass, momentum, and energy in a viscous fluid. These governing equations have the following structure¹⁰:

$$\partial_t U + \nabla \cdot \vec{F}^c - \nabla \cdot \vec{F}^v = Q \quad \text{in } \Omega \in \mathbb{R}^3, t > 0 \quad (4)$$

where t is the time and Ω is the domain. U denotes the vector of state variables; $\vec{F}^c$ are the convective fluxes (inviscid fluxes); $\vec{F}^v$ are the viscous fluxes; and Q is a source term.

Eqs. (5) and (6) give a brief description of the physics involved for each component of the governing equations:

$$U = (\rho, \rho v_1, \rho v_2, \rho v_3, \rho E)^T \quad (5)$$

$$\vec{F}^c = \begin{pmatrix} \rho v_i \\ \rho v_i v_1 + P \delta_{i1} \\ \rho v_i v_2 + P \delta_{i2} \\ \rho v_i v_3 + P \delta_{i3} \\ \rho v_i E \end{pmatrix}, \quad \vec{F}^v = \begin{pmatrix} 0 \\ \tau_{i1} \\ \tau_{i2} \\ \tau_{i3} \\ v_j \tau_{ij} + \mu_{tot}^* C_p \partial_i T \end{pmatrix}, \quad i=1 \dots 3 \quad (6)$$

where ρ is the density; P is the static pressure; E is the total energy per unit mass; H is the fluid enthalpy; and $\vec{v} = (v_1, v_2, v_3) \in \mathbb{R}^3$ is the flow velocity vector in a Cartesian coordinate system. In these formulations, the static pressure P could be derived from the state equation with the assumption of an ideal gas.

$$P = \rho(\gamma - 1) \left[E - \frac{1}{2} (v_1^2 + v_2^2 + v_3^2) \right] \quad (7)$$

$$T = \frac{P}{R\rho}, \quad C_p = \frac{\gamma R}{(\gamma - 1)} \quad (8)$$

where T is the temperature; γ is the specific heat ratio; and R is a gas constant. C_p is the specific heat at a constant pressure, and δ_{ij} is the Kronecker delta function. For the viscous flux term of Eq. (6), the viscous stresses can be written as Eq. (9). The viscosity is defined by Stokes' assumption.

$$\tau_{ij} = \mu_{tot} (\partial_j v_i + \partial_i v_j - \frac{2}{3} \delta_{ij} \nabla \cdot \vec{v}) \quad (9)$$

$$\mu_{tot} = \mu_{dyn} + \mu_{tur}, \quad \mu_{tot}^* = \frac{\mu_{dyn}}{Pr_d} + \frac{\mu_{tur}}{Pr_t} \quad (10)$$

The dynamic viscosity (molecular laminar viscosity) μ_{dyn} is derived using Sutherland's Law¹¹, and the turbulent viscosity μ_{tur} is calculated using a turbulence model. Pr_d and Pr_t are the Prandtl numbers for laminar and turbulent flows, which have values set at 0.72 and 0.9, respectively. In the present paper, we use the one-equation turbulence model of Spalart-Allmaras (SA) to compute turbulent viscosity μ_{tur} .

The spatial terms of the governing equations are discretized using the cell-vertex (node) based Finite Volume Method (FVM) with a median-dual control volume technique. The inviscid and viscous flux terms are evaluated using the central difference scheme, which is second-order accurate in space. For a high-resolution solution, we use the Jameson-Schmidt-Turkel (JST) scheme as an artificial viscosity term¹². To obtain a steady-state solution, we perform pseudo-time integration using the Lower-Upper Symmetric Gauss Seidel (LU-SGS)¹³ scheme. The SA turbulent model is used to consider turbulent eddies.

In this study, the flow around the multiple wing-sails has a very low speed. In this velocity region, it is important to consider the viscous effect for a more accurate flow solution. Because of the viscous effect on the wall, a boundary layer flow is developed and has to be accurately captured. Correspondingly, a high-resolution mesh is required near the wall with the value of y^+ close to unity. In the unstructured grid methodology, mixed grids (2D: quadrangle + triangle, 3D: hexahedron + tetrahedron) are generally utilized for the viscous computation. A mixed grid of quadrangles and triangles is used in the two-dimensional mesh, and a combination of hexahedrons and tetrahedrons is used for the three-dimensional mesh topology. The mixed grid topology for our multiple-wing-sail configuration is shown in Fig. 4. The size of the computational domain is $25 \times 10 \times 5$ times the chord length.

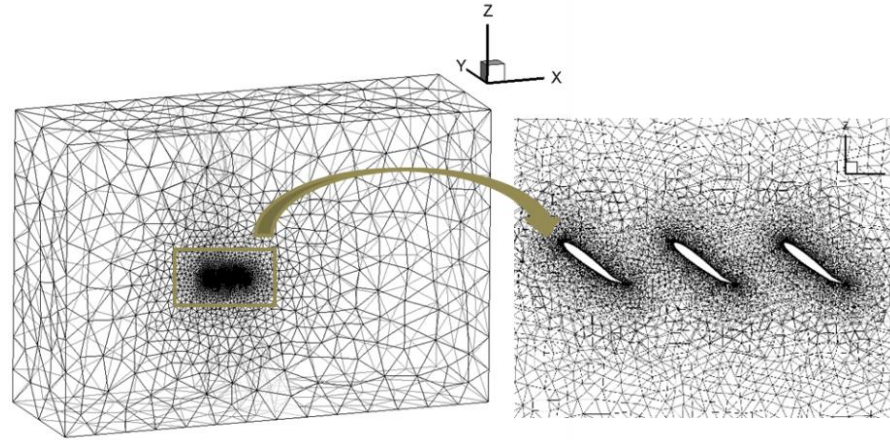


Fig. 4 Computational domain of mixed grid system around multiple wing-sails

In order to consider the viscosity effects, a no-slip boundary condition is used as a wall boundary condition on the wing-sail surface. A nonreflecting, far-field boundary condition using Riemann invariants is imposed on the far-field boundary condition, and the symmetry condition is used for the root plane of the wing-sails. The far-field boundary condition prevents reflected disturbances from coming back within the boundaries. In addition, because of the symmetry plane boundary condition, there is no flow across the boundary at the root plane of the wing-sails.

IV. Design Optimization Framework

In addition to the aerodynamic analysis of the multiple wing-sails, we developed a design optimization framework, which is tightly coupled with the high-fidelity flow analysis and gradient-free optimization algorithms. In this section, we explain two components of the current design optimization framework: Kriging surrogate model and GA. We also introduce a fully automated design optimization framework that integrates the geometry kernel, mesh generation and deformation, flow solution, and optimization algorithm.

A. Kriging surrogate model

Gradient-free optimization algorithms such as the GA are known to be easy and robust methods for design optimization. However, the objective function should be evaluated for every member in the population at each generation. If we have to perform a high-fidelity CFD computation for the function evaluation, it is practically impossible to apply GA directly to a design problem, because a single function evaluation may take a couple of hours, even with parallel computation. For this reason, a surrogate model or response surface method is introduced in order to substitute the function evaluation for the fit evaluations. The surrogate model, known as an approximate model, is an efficient way of alleviating the expensive computation burden. To construct the surrogate model, the function evaluation is required only for the initial experimental sample points, which are randomly selected in the design space. After the surrogate model is constructed, we can obtain any approximated value of the objective function without additional function evaluations.

There are two types of surrogate models. The polynomial-based regression type usually uses a least-squares method to emulate the trend of a real function. This regression technique focuses on estimating the relationships among function values and the trend of these values. Therefore, this approach is very sensitive to the distribution of data and does not guarantee that the resulting regression surface passes through all the sample data points¹⁴. On the other hand, another surrogate model method uses a data-fitting technique, which is known as interpolation. This interpolation-based surrogate model can construct a response surface that always passes through all the initial sampling points by using interpolation between the data points. For this reason, the interpolation-based surrogate model is known to be good for noisy and nonlinear functions such as CFD simulation. In this study, we use the Kriging surrogate model, which is popularly used in combination with the gradient-free methods. The Kriging surrogate model is defined as the sum of the mean and deviation terms, as shown in Eq. (11). The mean represents an approximate trend of the real function. The deviation term is a quantified value that is the difference between the real function and the approximated function. In other words, it is an error term.

$$y_{krig}(x) = f(x) + Z(x) \quad (11)$$

The mean term can be rewritten as a combination of the basis function and its coefficients, as shown in Eq. (12).

$$y_{krig}(x) = \sum_{j=1}^k \beta_j f_j(x) + Z(x) \quad (12)$$

The Kriging model can be classified as an ordinary, universal, or Taylor-Kriging model depending on the method used to formulate the mean term using polynomial equations. If we can predict the trend of the real function, the universal and Taylor-Kriging models are very useful approaches. However, in general engineering problems, because we do not know the exact trend for the real function, the ordinary Kriging model is mainly used. In the ordinary Kriging model, the polynomial term in Eq. (11) is considered to be a constant, namely, the 0th-order polynomial equation. Therefore, the ordinary Kriging model has the advantage of robust behavior for any problem^{15,16}.

The deviation term in Eq. (11) is represented by a normal distributed Gaussian process, which refers to the bias or uncertainty in the mean prediction. It is important to define the deviation term in the covariance relation, as in Eq. (13), which denotes the influence exerted by two points on each other.

$$\text{Cov}(Z(\bar{x}_1), Z(\bar{x}_2)) = \sigma_z^2 \mathbf{R}[\mathbf{R}(\bar{x}_1, \bar{x}_2)] \quad (13)$$

In Eq. (13), a correlation function R can be defined as a correlation matrix, which is used to interpolate between two points on a Kriging response surface. Therefore, how smoothly the Kriging surrogate model expresses any two points depends on what one is trying to define with the correlation function. Here, we consider a Gauss function as the correlation function. This function deals with the relation between two data sets by scaling their distance exponentially. Then, we apply an exponential correlation function, as shown in Eq. (14).

$$R(\bar{x}_1, \bar{x}_2) = \exp\left(\sum_{k=1}^n -\theta_k |\bar{x}_1^k - \bar{x}_2^k|^{p_k}\right) \quad (14)$$

As can be seen in Eq. (14), when the distance between $\bar{x}_1$ and $\bar{x}_2$ is further increased, the value of the correlation function approaches zero. In other words, the extent of influence between any two sample points is exponentially decreased. Furthermore, θ and p are parameters in the correlation function, and both are positive real variables. Because these parameters determine the rate of correlation of each independent variable, appropriate methods should be used for their accurate prediction. The magnitude of θ determines how fast the influence is weakened by the distance.

To construct the Kriging surrogate model, we need the initial sample points. These sample points can be extracted by using a Design of Experiment (DOE) technique. In this study, we employed a Latin Hypercube Sampling (LHS) method¹⁷, which is known to be suitable for Kriging surrogate model. In order to improve the accuracy and robustness of the Kriging surrogate model, we added adaptive sampling through a statistical Mean Square Error (MSE) estimation. After the Kriging surrogate model is constructed, it may be important to validate its accuracy. Therefore, the accuracy of this model should be verified by comparing the exact value from the function evaluation and the estimated value from the model.

B. Evolutionary algorithm

Evolutionary algorithms (EAs) are state-of-the-art techniques for optimization. EAs were inspired by the processes involved in the natural evolution of humans, including inheritance, mutation, selection, and crossover. Because EAs do not require gradient information, they are used widely for many problems in which gradient information is very difficult to compute. The GA, memetic algorithm, and neuroevolution are typical examples of EAs.

The Genetic Algorithm (GA) is one of the most popular EAs. When the design space is defined, the GA randomly selects a population set, which includes optimal solution candidates. Then, the function evaluation must be performed for every individual population. After all the function values are compared, the genetic operators select

better candidates as an inheritance set for the next generation. In the process of evolving to the next generation, the genetic operators apply mutation and crossover to evolve the optimum candidates, depending on the mutation probability specified by the user. Through the generations, the more evolved gene will survive among the populations. After many generations, the best one is chosen as the optimal solution. Commonly, these iterative generations are repeated until either a fixed number of generations is reached, or the optimal solution is found that satisfies the termination criteria.

In this study, Hybrid Genetic Algorithm (HGA) was used. In order to speed up the design procedure, the function values were evaluated using Kriging surrogate model. Fig. 5 shows the design procedure for multiple wing-sails. A brief description of the design procedure is as follows. To construct the Kriging surrogate model, we need initial sample points. The initial sample points for the response surface are randomly extracted by using Latin Hypercube Sampling (LHS) method, which is known to be suitable for the Kriging surrogate model. The surface geometries and corresponding computational meshes are automatically generated by the journaling systems. In this study, we employed the SU²_CFD solver, which is an unstructured mesh-based CFD flow solution method, to evaluate the aerodynamic performance of these initial sample points. Using a Kriging-based surrogate model and the derivative-free GA optimization method, we searched for the optimal candidate that maximizes the average thrust coefficient of multiple wing-sails. This optimal candidate was validated by making a comparison between the exact value from the function evaluation and the estimated value from Kriging surrogate model through SU²_CFD, which is a high-fidelity computational analysis. If differences between the estimated and validated performances are sufficiently small, the design optimization is terminated. Otherwise, the validated optimum is considered to be an additional point, and the response surface is reconstructed. This design procedure proceeds iteratively to find the optimal solution.

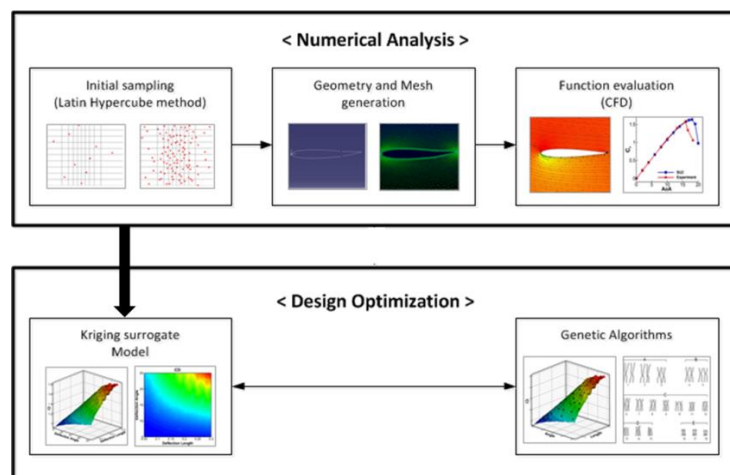


Fig. 5 Design framework for multiple wing-sails

V. Aerodynamic Characteristics of Wing-Sails

A. Validation of CFD flow solver

In order to examine the computational accuracy of the CFD flow solver, we performed a solver validation. By comparing the results of the numerical analysis and the experimental data, we confirmed the accuracy of the solver. As a validation model, we used the NACA0012 airfoil, which is a symmetrical and no-camber type of the 4-digit series of NACA airfoils. The aerodynamic experiment on the NACA0012 airfoil was conducted at $Re = 9.0 \text{ M}$ ¹⁸. The numerical analysis for NACA0012 was performed at $Re = 10.0 \text{ M}$, which is the design condition for multiple wing-sails. The Reynolds number based on the chord length of the wing-sail was 10^7 . The relative wind speed between the true wind speed and the ship cruise speed was 14 knots (7 m/s). For the solver validation, a mixed grid was generated, as shown in Fig. 6. The far field of the computational domain had a boundary, which was a radius of 20 times the airfoil chord length and a value of y^+ close to unity near the solid wall. The total numbers of nodes and elements in the computational grid are listed in Table 1.

In Fig. 7, although the drag coefficients predicted by the numerical analysis differed slightly from the experimental drag coefficients, the lift coefficients are considerably similar until an angle of attack of 16° . In addition, even though the numbers of drag counts have differences, the trends are very similar. Therefore, it is expected that the flow characteristics can be captured using the CFD flow solver. Thus, the results of the numerical analysis are sufficiently accurate.

Table 1 Numbers of nodes and elements in grid system

	Total number of nodes	Total number of elements
Single wing-sail in 2D analysis	17,650	26,620
Multiple wing-sails in 3D analysis	49,258	72,556
Multiple wing-sails in 3D analysis	1,217,710	6,541,622

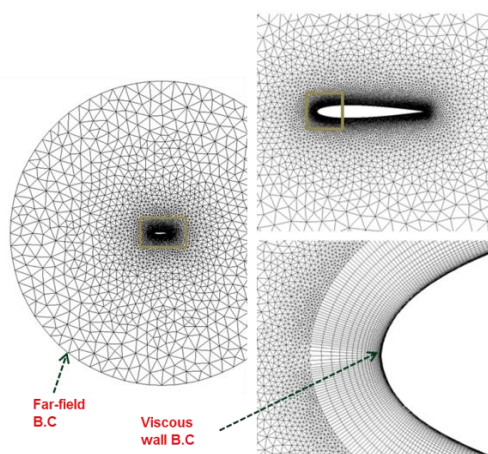


Fig. 6 Computational grid system around NACA0012

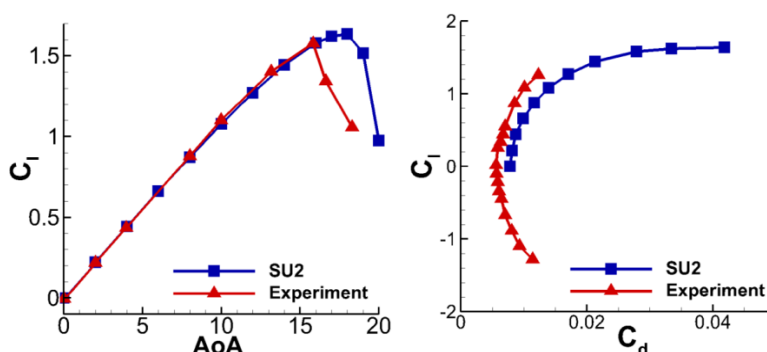


Fig. 7 C_l - α curve (left) and drag polar (right) of NACA0012

B. Aerodynamic characteristics of multiple wing-sails

Prior to investigating the effect of the flow interaction of multiple wing-sails, we carried out an aerodynamic analysis for a single wing-sail to reveal the effect of the flap. The angle of attack of a single wing-sail is allowed to vary from 9° to 21° in steps of 3° . As mentioned before, the single wing-sail is a rectangular wing with an aspect ratio of 2. The two-dimensional cross section shape is an NACA0012 airfoil with and without a trailing edge flap, which can be deflected up to an angle of 15° . The length of the flap is 20% of the chord length. In Fig. 8, we compare the values of the aerodynamic coefficients C_l and C_d between the single wing-sail cases with and without a flap. It can be seen that the single wing-sail with the flap has approximately 20~50% higher lift and drag coefficients

in comparison to the one without the flap. If we employ the trailing edge flap, the effective camber of the airfoil is increased, and the pressure gradient from the stagnation point to the suction peak point is increased. This results in increased velocity at the suction peak (low-pressure region on the upper surface), which causes a larger pressure difference between the upper and the lower surfaces of the wing-sail. For these reasons, the single wing-sail with a flap can have better aerodynamic performance than the one without a flap. Furthermore, we can observe from Fig. 8 that the stall angle of the wing-sail without the flap was about 15° , whereas that of the wing-sail with the flap was about 12° .

To investigate the effect of the flow interaction, which has a significant effect on the aerodynamic performance of multiple wing-sails, we performed a numerical aerodynamic analysis for multiple wing-sails by varying the wind direction from 15° to 165° in intervals of 15° . Even though the wind direction varied, the angle of attack of each wing-sail, which is a relative angle between the wing-sails and the wind direction, was fixed at 12° . The value of 12° was obtained by examining the C_l - α curve in Fig. 8. The maximum lift of the single wing-sail with a flap was obtained at that angle.

As indicated in Fig. 9, because of the camber effect of the flap, the thrust performance of multiple and single wing-sails with a flap is superior to those without a flap. Furthermore, we considered how much thrust is increased or decreased by the flow interaction in multiple wing-sails by comparing the results with the thrust of a single wing-sail. In the comparison of the thrust coefficients of a single wing-sail and multiple wing-sails, both with and without flaps, the thrust coefficients of the multiple wing-sails were smaller by about 21~43% (with flap case) and 24~37% (without flap case) than those of the single wing-sail. However, in the multiple wing-sails, the average thrust coefficient is divided by the total area of the wing-sails. Therefore, the total area should be considered to obtain the total thrust force. As a result, the total thrust force of multiple wing-sails is about six times that of a single wing-sail. From these results, we conclude that the flow interaction between the wing-sails results in decreased aerodynamic performance. As a result, the thrust coefficients of multiple wing-sails are lower than that of a single wing-sail at all wind directions. The mechanism of the flow interaction that causes the decreased aerodynamic performance in multiple wing-sails is as follows. If the wing-sail is set with a positive angle of attack, a stagnation point is located on the lower surface of the wing-sail instead of at the leading edge. Furthermore, a suction peak point, which is a very low pressure region, is located on the upper surface of the wing-sail. The front wing-sail, which is located further upstream in the wind direction, causes the stagnation point on the rear wing-sail to shift toward the leading edge; "the head effect." Then, the strength of the suction peak point of the rear wing-sails is considerably reduced. Because the pressure gradient from the stagnation point to the suction peak point is also decreased, the flow velocity is less accelerated (the suction peak velocities are reduced). For these reasons, on the upper surface of the rear wing-sail, the recovery adverse pressure gradient is also lower. Then, the rear wing-sails can be operated at a higher angle of attack without flow separation and stall. As a result of the above, although the each angle of attack of the multiple wing-sails is set to be the same, the effective angle of attack of the rear wing-sail is reduced because of the flow interaction effect¹⁹.

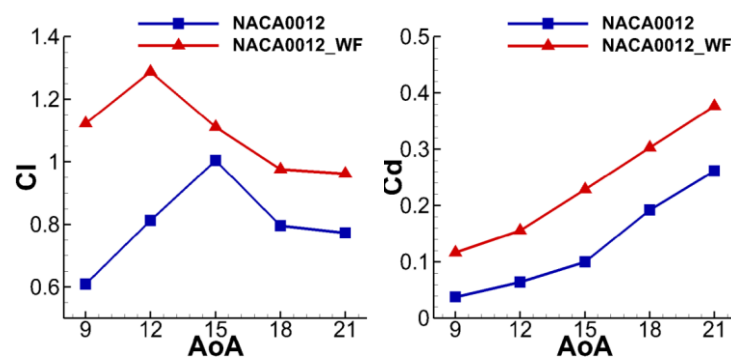


Fig. 8 Aerodynamic coefficient comparison between single wing-sails with and without flap

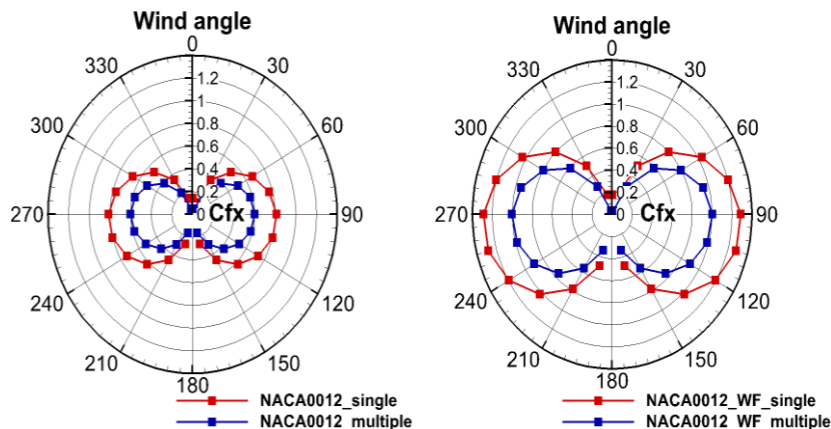


Fig. 9 C_{fx} comparison between single and multiple wing-sails without flap (left), and C_{fx} comparison between single and multiple wing-sails with flap (right)

VI. Design Applications and Results

In this section, we show the design optimization results for single and multiple wing-sails. For a single wing-sail, we constructed the exact response surfaces based on aerodynamic performances and selected the optimum solution visually. In contrast, the design optimization framework was used for multiple wing-sails under various wind direction conditions. In order to speed up the design procedure, we performed the design optimization in two dimensions (2D) and validated the design results in three dimensions (3D).

A. Design problem formulation

In the previous study, we considered only the angle of attack of each wing-sail as design variables^{5,6}. Therefore, the geometric deformation of the wing-sails was not considered. However, in this research, we found the optimum shape and operating condition for multiple (three) wing-sails. A total of nine design variables are employed, including the flap length, deflection angle, and angle of attack, in terms of the flap configuration and operating condition. The initial flap length and deflection angle were set at 20% of the chord length and 15°, respectively. In the previous study, we set the initial angle of attack at 12°. However, this initial angle of attack is too high to find a steady-state solution because it causes flow separation on the head wing-sail, which was the first wind-sail in the wind direction. Therefore, in this study, 8° and 9.5° were chosen as the initial angles of attack of each wing-sail. Fig. 10 shows a schematic diagram of the design variables for the design optimization of the multiple wing-sails. The angle of attack is the relative angle between the wing-sail direction and the wind direction. Constraints on the design variables, called design bounds, were set from +15° to -15° (angle of attack), from +5° to +25° (flap deflection angle), and from 0.05c to 0.3c (flap deflection length).

In a parameter study of a single wing-sail, because no flow interaction occurs, the angle of attack remains constant at 8°. In other words, we consider that the design variables are only the deflection angle and flap length. Therefore, we are able to construct the exact response surface with a 30 × 30 resolution, which is an output, corresponding to two input variables (deflection angle and length). After the exact surfaces were organized using 900 function evaluations, the optimum setup of the flap geometry was selected visually. On the other hand, for the design optimization of multiple wing-sails, we considered a total of nine design variables (three design variables for each wing-sail × three wing-sails). Therefore, we employed the design framework, which tightly coupled the high-fidelity CFD and gradient-free optimization algorithms.

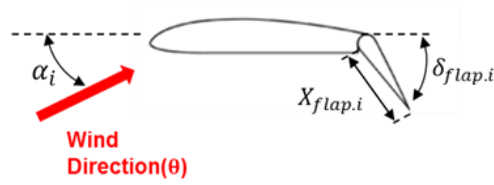


Fig. 10 Design variables for multiple wing-sails

$$\begin{aligned}
 &\text{maximize } I = C_{fx}(\bar{x}_i, \bar{w}_i) \\
 &\text{by varying } \bar{x}_i(\bar{\delta}_i, \bar{f}_i) \in \mathbb{R} \quad (i = 1, 2, 3) \\
 &\quad \quad \quad \bar{w}_i(\alpha_i) \in \mathbb{R} \quad (i = 1, 2, 3) \\
 &\text{such that } \alpha_l \leq \alpha \leq \alpha_u \\
 &\quad \quad \quad \delta_l \leq \delta \leq \delta_u \\
 &\quad \quad \quad X_{flap,l} \leq X_{flap} \leq X_{flap,u} \\
 &\quad \quad \quad C_{fx,opt} \geq C_{fx,base}
 \end{aligned}$$

B. Parameter study using single wing-sail

Using 900 function evaluations, the two exact response surfaces for the lift and drag were constructed, which are shown in Fig. 11. We chose a flap deflection length of 0.2914c, and a flap deflection angle of 24.3103° as the optimum for a single wing-sail, which generated the maximum aerodynamic performance. In Fig. 11, the airfoil with the larger flap length and angle (red circle) has a stronger suction peak on the upper surface of the wing-sail compared to those of the airfoil with the smaller flap length and angle (black circle). These have a larger “camber effect,” which makes the lift force more effective. Based on this result, it can be inferred that the design for multiple wing-sails points in the direction of a larger flap length and angle.

The thrust is the resultant force of the lift and drag forces. It can be calculated using Eq. (1) along the wind direction, and the results are listed in Table 2. Because no flow interaction occurs with a single wing-sail, the lift and drag coefficients remain constant with respect to the wind direction. However, the thrust coefficient in a wind direction of 135° is larger than the thrust coefficients in the other wind directions. In addition, the thrust coefficient is maximized in the wind direction of 90°. This can be proved using Eq. (1). When the wind direction is varied from 0° to 90°, the sign of the drag is positive. It generates a negative thrust, which is the opposite to that of the lift. Thus, the resultant force becomes smaller. When the wind direction is varied from 90° to 180°, both the lift and the drag have positive signs, which cause the resultant force to become larger. In a wind direction of 90°, the drag does not generate any thrust, and only the lift contributes to the thrust. This is why the maximal thrust occurs in a wind direction of 90°.

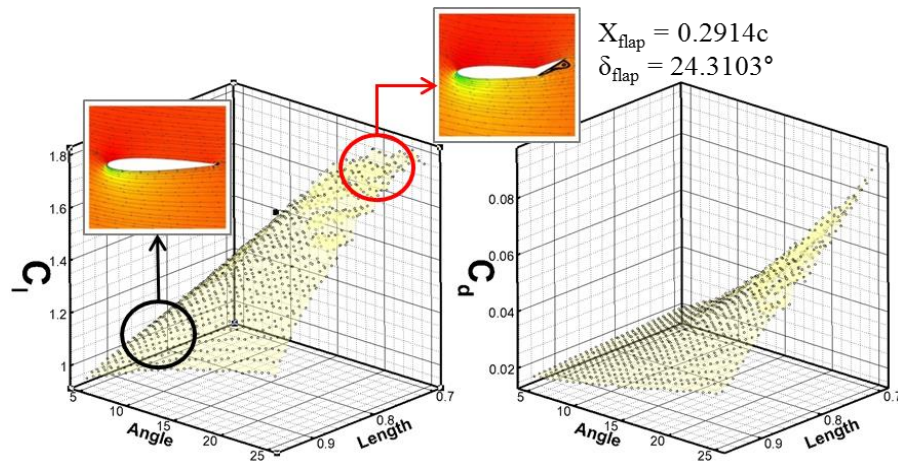


Fig. 11 Exact response surfaces of C_l (left) and C_d (right) for single wing-sail

Table 2 Parameter study results for single wing-sail

	C_l	C_d	C_{fx}
$\theta = 45^\circ$	1.8211	0.0829	1.2291
$\theta = 90^\circ$	1.8211	0.0829	1.8211
$\theta = 135^\circ$	1.8211	0.0829	1.3463

C. Design results of multiple wing-sails at angle of attack of 8°

As mentioned before, for the design optimization of multiple wing-sails, we considered a total of nine design variables, which included the angle of attack, flap deflection length, and angle of each wing-sail. Therefore, we employed the gradient-free optimization algorithms and Kriging surrogate model. A total of 90 sample points were estimated for constructing the Kriging response surfaces for each wind direction case. In addition, some adaptive sampling procedures and the regeneration of the response surface were performed to improve the accuracy of the Kriging response surface. To evaluate the sample points, which were extracted by using LHS method, we generated a computational grid with 49,258 nodes and 72,556 elements.

1. Design with respect to varying wind direction

The results of the design optimization are listed in Table 3. In Table 3, the variation of the angle of attack ($\Delta \alpha$) refers to the amount of change from the initial angle of attack to the optimum angle of attack. The sign of the changing angle follows the “right hand rule.” If the sign is positive, the angle of attack becomes larger than the initial angle of attack, whereas the angle of attack becomes smaller with a negative sign.

- Case of wind direction $\theta = 45^\circ$

As seen in Table 3, the three types of design variables were gradually increased, which gradually improved the aerodynamic performances. It is notable that the angle of attack of wing-sail #1 was decreased, whereas those of wing-sails #2 and #3 were increased. We confirmed the header effect, as discussed previously. On the baseline results, the stagnation point gradually moved toward the leading edge of each wing-sail because of the flow interaction. Then, the effective angles of attack were decreased. This phenomenon contributed to the strength variation of the suction peak and the flow separation. A larger effective angle of attack led to a stronger suction peak and flow separation. In a comparison between the baseline and the design results, the strength of the suction peak on wing-sail #1 was decreased slightly. This is because the angle of attack of wing-sail #1 was reduced to prevent flow separation at the trailing edge flap. Therefore, the flow separation almost disappeared. This was also

caused by the reduced flap length and angle. As a result, the design variables of the head wing-sail, which was the first wind-sail in the wind direction, were reduced (decreased angle of attack and deflection angle, and shortened flap length) to prevent the flow separation. On the other hand, the design variables of wing-sails #2 and #3 were gradually increased (increased angle of attack and deflection angle, and lengthened flap length) to improve the aerodynamic performance. For these reasons, the strengths of the suction peaks were increased. In addition, the pressure differences between the upper and lower surfaces of wing-sails #2 and #3 were significantly increased, as shown in Fig. 12. Fig. 12 shows the pressure contour, streamline, and surface pressure coefficient distribution around the wing-sails. The “blue circle” shows the stagnation region of each wing-sail, and the “red circle” shows the flow separation on the upper surface of the flap of each wing-sail. As can be seen in Fig. 12, in a comparison of the baseline and optimized flow fields, the stagnation point shifted from the lower surface of the wing-sail to the leading edge, and the flow separation disappeared.

- Case of wind direction $\theta = 90^\circ$

In Table 3, there are no trends for the same types of design variables. It is thought that relatively less flow interaction occurred in a wind direction of 90° . In the previous study, the smallest flow interaction was observed in a wind direction of 90° . The overall angles of attack of the wing-sails were increased to enhance the aerodynamic performance. For wing-sail #2, the flap deflection length is very small in comparison with the other wing-sails. Therefore, to compensate for the shorter deflection length, the angle of attack and deflection angle of wing-sail #2 are considerably greater than those of the other wing-sails. In a wind direction of 90° , because little flow interaction occurred, each wing-sail was optimized independently to find the optimal setup of the design variables. As a result, we obtained an improvement in the aerodynamic performances of the wing-sails that was caused by an increased angle of attack.

- Case of wind direction $\theta = 135^\circ$

As shown in Table 3, the design trends were the reverse to the trends in the case of a wind direction of 45° , because wing-sail #3 is the head wing-sail instead of wing-sail #1. However, the angle of attack of the head wing-sail (wing-sail #3) is slightly increased. Thus, it is thought that this case has less flow interaction than that with a wind direction of 45° . As a result, the angle of attack, deflection length, and angle of attack of the rear wing-sail, which is located further downstream, were gradually increased to compensate for the header effect.

Table 3 Design optimization results for multiple wing-sails at initial angle of attack of 8°

	$\theta = 45^\circ$			$\theta = 90^\circ$			$\theta = 135^\circ$		
	Sail #1	Sail #2	Sail #3	Sail #1	Sail #2	Sail #3	Sail #1	Sail #2	Sail #3
$\Delta \alpha (^\circ)$	-2.7300	2.3379	8.0323	0.7579	7.5028	4.4230	4.6671	4.4461	0.9149
$X_{\text{flap}} (\% \text{ of } c)$	0.1759	0.2136	0.2751	0.2169	0.0986	0.2421	0.2183	0.2096	0.1695
$\delta_{\text{flap}} (^\circ)$	9.9131	12.453	17.990	13.549	15.184	10.371	17.386	8.3102	8.6778

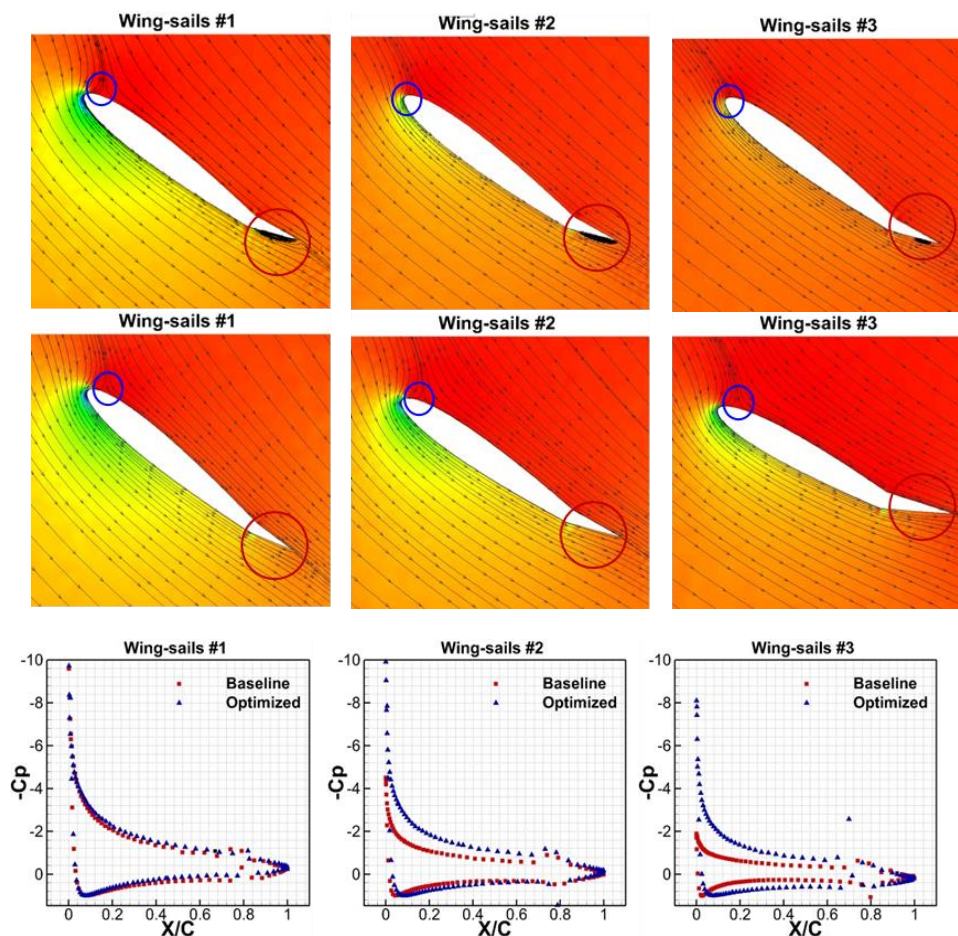


Fig. 12 Comparison of flow fields at 45° wind direction using two-dimensional analysis: pressure contour and streamline around baseline (top) and optimized (center) wind-sails, and surface pressure coefficient distribution of both baseline and design results (bottom)

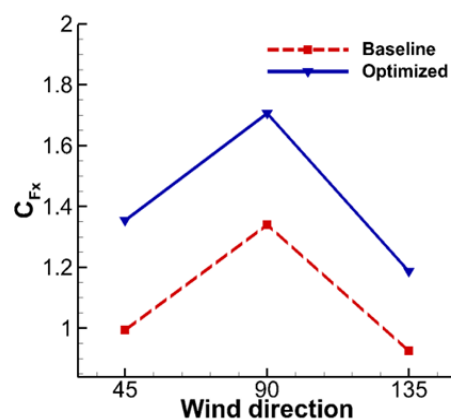


Fig. 13 C_{fx} comparison between baseline and optimized results using two-dimensional analysis

For a comparison of the three cases, the averaged thrusts of the baseline and optimized results are plotted in Fig. 13 with respect to the varying wind direction. In addition, the values of each averaged thrust are listed in Table 4. As we described, in a wind direction of 90°, the maximum averaged thrust was obtained in both the baseline and

optimized results, and the improvements in the averaged thrust are similar in the wind directions of 90° and 135°. The relative improvements are the largest in the case of a wind direction of 45°, which caused very strong flow interactions. The overall improvements in the averaged thrust were 14~22%, which are very successful design results.

Table 4 C_{fx} improvement in multiple wing-sails using two-dimensional analysis

	$\theta = 45^\circ$		$\theta = 90^\circ$		$\theta = 135^\circ$	
	C_{fx_ave}	Improve- ment (%)	C_{fx_ave}	Improve- ment (%)	C_{fx_ave}	Improve- ment (%)
Baseline	1.1081	-	1.4927	-	1.0362	-
Optimized	1.3542	22.21	1.7057	14.27	1.1860	14.45

2. Validation of design results using three-dimensional CFD analysis

The design results were based on a two-dimensional analysis. We carried out a three-dimensional numerical analysis to determine whether the design results achieved with the two-dimensional assumption were preserved. We generated a full three-dimension computational grid using the same mixed grid topology. The numbers of nodes and cells in the grid were 1,217,710 and 6,541,622, respectively, as shown in Table 1. The same flow conditions as used in the two-dimensional analysis were used in the three-dimensional analysis.

In aerodynamic characteristics of aircraft, a geometrical finite wing causes a “3D effect,” in other words, “a wing-tip vortex.” This is driven by the pressure difference between the upper and lower surfaces of the wing. In general, the fluid moves from a high-pressure region to a low-pressure region. If the wing has a positive angle of attack, the high-pressure region is distributed on the lower surface, whereas the low-pressure region is distributed on the upper surface of the wing. Therefore, the fluid moves from the lower surface to the upper surface at the wing tip. This phenomenon generates the vortex, which results in a downwash flow. As a result, the effective angle of attack is reduced around the wing tip because of the wing-tip vortex, which results in decreased lift and increased drag. Because this flow phenomenon occurs in actual wing-sails, it should be considered.

In Fig. 14, the “red circle” shows the suction peak region on the upper surface of the wing-sail. The top of the figure shows the flow fields of the baseline, whereas the center of the figure shows the flow fields of the design result. The overall flow characteristics around the three-dimensional wing-sail are similar to those for the two-dimensional wing-sail. However, the pressure differences between the upper and lower surfaces of each wing-sail are relatively smaller than the flow fields in Fig. 12. In addition, no flow separations occurred on the flap surfaces in three dimensional analyses. This means that even though the angle of attack was set to be the same (8°) in the two- and three-dimensional analyses, the flow characteristics around the multiple wing-sails were slightly different because of the 3D effect. To investigate whether the 2D improvements were valid in three dimensions, we plot the averaged thrust coefficients along the wind direction for both the baseline and the optimized results in Fig. 15. The averaged thrusts were improved in all wind directions. The improvements in the averaged thrusts can be confirmed in Table. 5. The overall improvements are 10~17%, which are smaller than the improvements in the 2D design optimization. The differences between the improvements in the 2D and 3D design optimizations were 2~5%, which is very predictable considering the 3D effect. Therefore, we concluded that the developed design optimization framework for multiple wing-sails, which was based on the 2D design optimization, could be utilized for 3D design optimization. Furthermore, we confirmed that the design optimization trends in two dimensions were still preserved in three dimensions.

Table. 5 C_{fx} improvement in multiple wing-sails using three-dimensional analysis

	$\theta = 45^\circ$		$\theta = 90^\circ$		$\theta = 135^\circ$	
	C_{fx_ave}	Improve- ment (%)	C_{fx_ave}	Improve- ment (%)	C_{fx_ave}	Improve- ment (%)
Baseline	0.5462	-	0.9525	-	0.7372	-
Optimized	0.6430	17.72	1.0676	12.08	0.8110	10.02

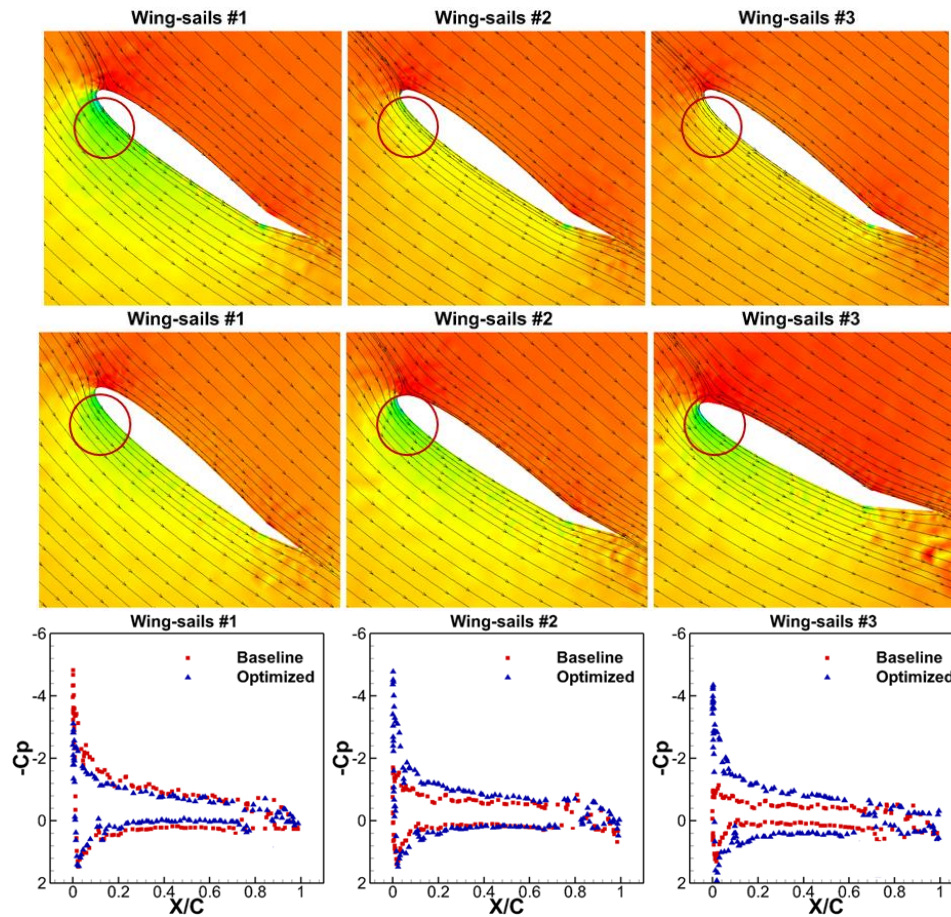


Fig. 14 Comparison of flow fields in 45° wind direction using three-dimensional analysis at $Y/C = 5\%$: pressure contour and streamline around baseline (top) and optimized (center) wind-sails, and surface pressure coefficient distributions of both baseline and design results (bottom)

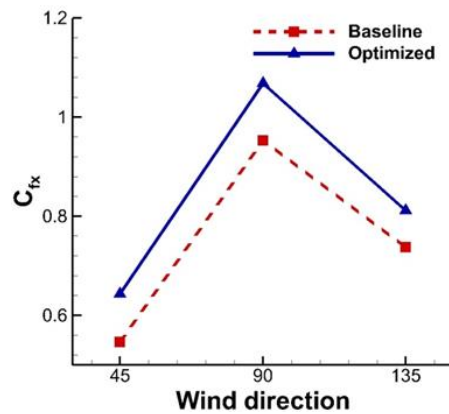


Fig. 15 C_{fx} comparison between baseline and optimized results using three-dimensional analysis

D. Design results for multiple wing-sails at angle of attack of 9.5°

1. Design with respect to varying wind directions

- Case of wind direction $\theta = 45^\circ$

In this study, we also performed the design optimization for multiple wing-sails to improve the average-thrust performance in the case of an initial angle of attack of 9.5° . This operating condition is closer to the stall condition, which results in the flow separation. The part of the flow that separates from the boundary layer is called the separation bubble flow and is a recirculating flow. This flow cannot move along the streamline. After the flow goes farther downstream, it eventually attaches to the wall again, which is called the reattached flow. As can be seen in Fig. 16, unlike the case of an angle of attack of 8° , a separation bubble occurs on the upper surface of wing-sail #1. In a comparison between the baseline and optimized results, by decreasing the angle of attack of wing-sail #1, the region of the separation bubble was considerably reduced. However, in the case of wing-sails #2 and #3, to compensate for the angle of attack reduction by the flow interaction, each angle of attack was increased. As a result, as compared with the baseline, the aerodynamic performance of all the wing-sails was improved. This can be confirmed from the surface pressure distribution graph in Fig. 16.

- Case of wind direction $\theta = 90^\circ$

Similar to the case of an initial angle of attack of 8° , there were no trends for the same types of design variables. It is thought that relatively less flow interaction occurred in this case. Therefore, because the effect of the flow interaction was smaller than that in the other wind directions, each wing-sail was optimized to improve its aerodynamic performance. Furthermore, because the wing-sails in the wind direction of 90° have little flow interaction with each other as compared to those in the other wind directions, the maximum thrust of the wing-sails is generated in a wind direction of 90° .

- Case of wind direction $\theta = 135^\circ$

In this case, the design trends were very similar to the trends in a wind direction of 135° for an initial angle of attack of 8° . However, unlike the case of an angle of attack of 8° , the angle of attack of the head wing-sail (wing-sail #3) was reduced to prevent the flow separation. It is thought that this initial angle of attack is too high. The angles of attack, flap lengths, and deflection angles of wing-sails #1 and #2 were gradually increased to improve the aerodynamic performance of the multiple wing-sails in consideration of their flow interaction.

Table. 6 Design optimization results for multiple wing-sails at initial angle of attack of 9.5°

	$\theta = 45^\circ$			$\theta = 90^\circ$			$\theta = 135^\circ$		
	Sail #1	Sail #2	Sail #3	Sail #1	Sail #2	Sail #3	Sail #1	Sail #2	Sail #3
$\Delta \alpha (^\circ)$	-2.0052	2.5522	6.1504	1.0898	5.7665	4.1063	7.9377	2.6348	-1.4463
$X_{\text{flap}} (\% \text{ of } c)$	0.1630	0.1866	0.2216	0.2117	0.1999	0.2191	0.1908	0.1994	0.2181
$\delta_{\text{flap}} (^\circ)$	10.299	11.436	13.707	13.303	12.351	10.936	11.460	11.856	10.967

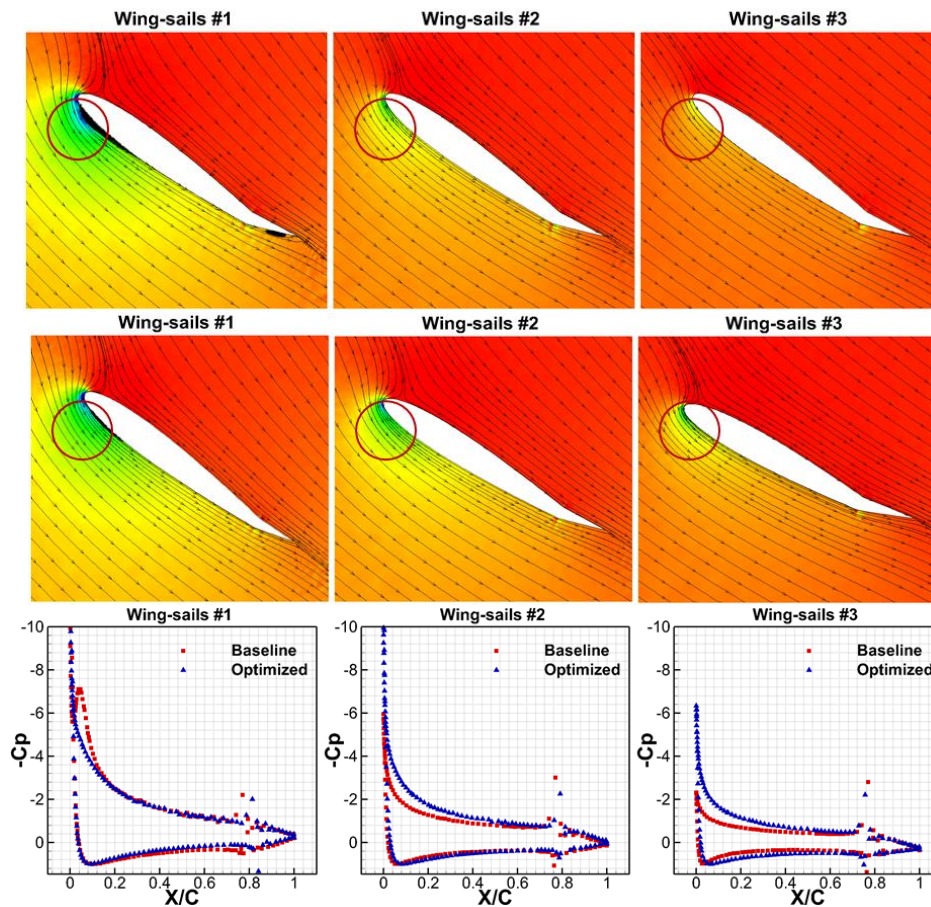


Fig. 16 Comparison of flow fields in 45° wind direction using two-dimensional analysis: pressure contour and streamline around baseline (top) and optimized (center) wind-sails, and surface pressure coefficient distribution of both baseline and design results (bottom)

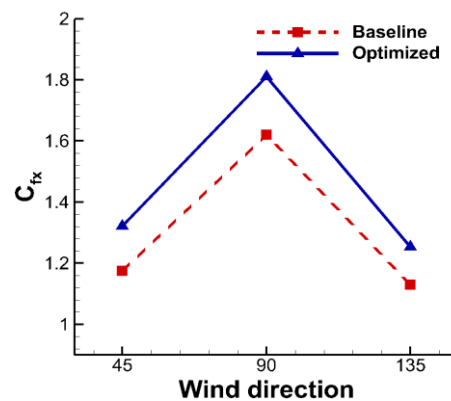


Fig. 17 C_{fx} comparison between baseline and optimized results using two-dimensional analysis

In summary, in the wind directions of 45° and 135° , the angle of attack of the front wing-sail, which is located further upstream in the wind direction, was reduced and the other angles of attack of the wing-sails were gradually increased. On the other hand, in a wind direction of 90° , all of the angles of attack of the wing-sails were increased. In the wind directions of 45° and 135° , because a strong flow interaction occurred and the initial angle of attack was too high, the angle of attack of the head wing-sail (when the wind direction was 45° , the head wing-sail was wing-sail #1, and when it was 135° , the head wing-sail was wing-sail #3) was reduced slightly to prevent flow separation, which results in a reduction in the aerodynamic performance of the wing-sail. On the other hand, the angles of attack, flap lengths, and deflection angles of the other wing-sails were gradually increased to improve the aerodynamic performance.

In a wind direction of 90° , because a weak flow interaction occurred, we could not confirm a distinct design trend. Therefore, the design variables of each wing-sail were varied independently to improve the aerodynamic performance. We could conclude that the maximum thrust of the wing-sails was generated in a wind direction of 90° , because the wing-sails in a wind direction of 90° had little flow interaction with each other as compared to those in the other wind directions.

As can be seen in Fig. 17, we were able to obtain an overall improvement in the averaged thrust of approximately 10~12%, depending on the wind direction and corresponding detailed flow characteristics.

Table. 7 C_{fx} improvement in multiple wing-sails using two-dimensional analysis

	$\theta = 45^\circ$		$\theta = 90^\circ$		$\theta = 135^\circ$	
	C_{fx}	Improve- ment (%)	C_{fx}	Improve- ment (%)	C_{fx}	Improve- ment (%)
Baseline	1.1752	-	1.6198	-	1.1303	-
Optimized	1.3219	12.49	1.8104	11.77	1.2528	10.85

2. Validation of design results using three-dimensional analysis

A high-fidelity CFD analysis was carried out to validate the design results in three dimensions. The validation results show that the average thrust performance of optimized multiple wing-sails was improved in all wind directions in comparison with the baseline multiple wing-sail. As can be seen in Table. 8, we were able to obtain an actual improvement of 9~11%, depending on the wind direction.

As shown in Fig. 18, we plotted the streamline and pressure contour around multiple wing-sails at a y/c of 5%. A comparison of the baseline and optimized results shows that the strength of the suction peak point of each wing-sail was increased, except for wing-sail #1. In the case of the head wing-sail (wing-sail #1), which is located at first in the wind direction, the angle of attack was decreased to prevent flow separation in two dimensions. However, as can be seen in Fig. 18, flow separation did not occur for any of the wing-sails because of the 3D effect, which is driven by the pressure difference between the upper and lower surfaces of the wing. Therefore, the aerodynamic performance of the optimized head wing-sail was worse than the baseline. However, in the case of wing-sails #2 and #3, the strength of the suction peak was distinctly increased, which was caused by increasing the angle of attack. Then, the flow was more accelerated from the stagnation point to the suction peak, which resulted in increased lift and drag performances. This can also be confirmed from the surface pressure distribution graph in Fig. 18.

Table. 8 C_{fx} improvement in multiple wing-sails using three-dimensional analysis

	$\theta = 45^\circ$		$\theta = 90^\circ$		$\theta = 135^\circ$	
	C_{fx}	Improve- ment (%)	C_{fx}	Improve- ment (%)	C_{fx}	Improve- ment (%)
Baseline	0.5900	-	1.0333	-	0.8080	-
Optimized	0.6452	9.36	1.1481	11.11	0.8815	9.10

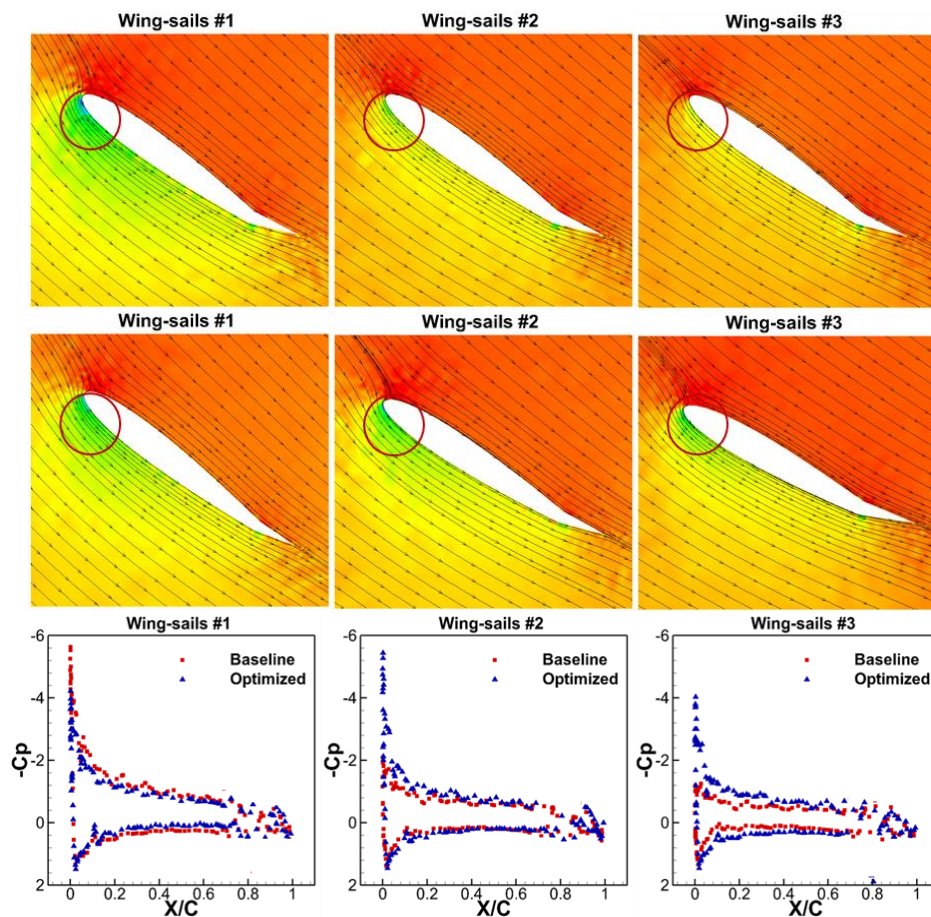


Fig. 18 Comparison of flow field in 45° wind direction using three-dimensional analysis at $Y/C = 5\%$: pressure contour and streamline around baseline (top) and optimized (center) wind-sails, and surface pressure coefficient distribution of both baseline and design results (bottom)

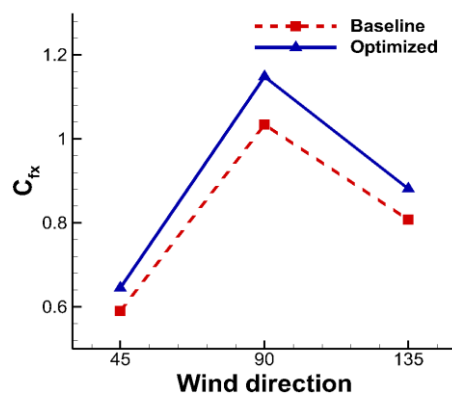


Fig. 19 C_{fx} comparison between baseline and optimized results using three-dimensional analysis

VII. Conclusion and Future work

In this research, we focused on an aerodynamic analysis and design optimization of multiple wing-sails. For the aerodynamic analysis, we solved the two/three-dimensional (2D/3D) compressible Navier-Stokes governing equation with a Spalart-Allmaras turbulence model. To more exactly predict the viscous flow, we used a mixed grid topology. Using the gradient-free optimization method of Genetic Algorithms (GAs) and Kriging Surrogate model method, we found the optimal setup for multiple wing-sails that maximizes the thrust performance. In a previous study, we only dealt with the angle of attack of each wing-sail as design variables. Namely, the shape deformation of the wing-sails was not considered. However, in this study, we set the angle of attack, flap deflection angle, and deflection length of individual wing-sails as design variables. The total number of design variables was nine.

Until now, studies on wing-sails depended on aerodynamic analyses using wind tunnel experiments. In this study, we developed a design framework for multiple wing-sails to improve the aerodynamic performance that tightly coupled high-fidelity computational fluid dynamics (CFD) and gradient-free design optimization algorithms.

Prior to the design optimization, we performed an aerodynamic analysis to reveal the effect of flow interaction and the flow characteristic around multiple wing-sails. For the numerical analysis, we used the SU²-CFD solver. We investigated the effect of the flap of a wing-sail. The results showed that a wing-sail with a flap has about 20~50% larger lift and drag coefficients than a wing-sail without a flap with respect to the angle of attack. In multiple wing-sails, the effect of their interaction should be considered so as to estimate their thrust. In a comparison between a single wing-sail and multiple wing-sails, the thrust coefficients of the multiple wing-sails were worse than those of a single wing-sail by about 21~43%, depending on the wind direction, because of the flow interaction, which decreased the effective angle of attack of the wing-sails located further downstream in the wind direction. Furthermore, compared to the other wind directions, the wing-sails in a wind direction of 90° showed little flow interaction with each other and had the best aerodynamic performance.

To confirm the relation of the design variables, which were the flap deflection length and angle in two dimensions, we constructed an exact response surface for the aerodynamic performance of a single wing-sail. We performed the numerical analysis 900 times. Through the exact response surface, it was possible to select the optimum point. The optimal setup for the two-dimension single wing-sail was a flap length of 0.2914c and a flap angle of 24.3103° at a fixed angle of attack. Furthermore, using the developed design framework, optimal design variables were obtained that maximize the averaged thrust performance of multiple wing-sails with respect to the wind direction. We carried out design optimization for initial angles of attack of 8° and 9.5°. With multiple wing-sails, the 9.5° angle of attack was closer to the stall condition. As a result, we acquired thrust increases of approximately 14~22% and 10~12%, respectively, in two dimensions. A high-fidelity CFD analysis was carried out for complete three-dimensional wing-sails to determine whether the thrust improvement achieved from the two-dimensional assumption was preserved. The results showed that the thrust performance of the multiple wing-sails was improved by 10~17% and 9~11%, respectively, in three dimensions. Although the thrust increments were reduced because of the 3D effect, these were still good. Thus, it was concluded that the design optimizations were successful when using the design framework.

We plan to conduct advanced research on wing-sails that can be applied practically to an actual bulk ship using a stability analysis and structural analysis. Furthermore, future studies will consider various aspects such as the interaction between the wing-sails and the ship and the optimization of wing-sails in three dimensions using efficient optimization methods. Establishing a wing-sail system based on the results of these studies is expected to contribute to the development of more efficient eco-friendly ships.

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References

- ¹Kazuyuki Ouchi, Kiyoshi Uzawa, Akihiro Kanai, "Huge Hard Wing Sails for the Propulsor of Next Generation Sailing Vessels", Second International Symposium on Marine Propulsor, 2011.
- ²Toshifumi Fujiwara, Koichi Hirata, Michio Ueno, Tadashi Nimura, "On Aerodynamic Characteristics of a Hybrid-sail with Square Soft Sails", Proceedings of the 13th International Offshore and Polar Engineering Conference, 2003.
- ³Toshifumi Fujiwara, Grant E. Hearn, Fumitoshi Kitamura and Michio Ueno, "Sail-sail and sail-hull interaction effects of hybrid-sail assisted bulk carrier", Journal of Marine Science and Technology 10:82-95, 2005.
- ⁴Takuji Nakashima, Yasunori Nihei, Yoshihiro Yamashita, Yasuaki Doi, "A Basic Study on Aerodynamic Interaction of Rigid Wing-sails for a Next Generation Wind Driven Vessel", First World NAOE Forum, Technical Challenges for Ship Efficiency toward 2020, No. 7, 2009.
- ⁵Hakjin Lee, Yeongmin Jo, Seongim Choi, Jongoh Kwon, and Sungmok Ahn, "Aerodynamic Analysis and Optimization of Wing-sails", Korean Society for Computational Fluids Engineering, 2012
- ⁶Hakjin Lee, Yeongmin Jo, Seongim Choi, Jongoh Kwon, and Sungmok Ahn, "Aerodynamic Analysis and Optimization of Wing-sails", Proceeding of the PRADS, 2013
- ⁷Ronald O' Rourke, "Navy Ship Propulsion Technologies: Options for Reducing Oil Use – Background for Congress", Congressional Research Service Report for Congress, 2006.
- ⁸Takuji Nakashima, Yoshihiro Yamashita, Yasunori Nihei and Quao Li, "A Basic Study for Propulsive Performance Prediction of a Cascade of Wing Sails Considering Their Aerodynamic Interaction", Proceeding of the Twenty-first (2011) International Offshore and Polar Engineering Conference (ISOPE), 2011
- ⁹Stanford University Unstructured (SU²), "su2.stanford.edu"
- ¹⁰Francisco Palacios, Michael R. Colonno, Aniket C. Aranake, Alejandro Compos, Sean R. Copeland, Thomas D. Economon, Amita K. Lonkar, Trent W. Lukaczyk, Thomas W. R. Taylor and Juan J. Alonso, 51st AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition, AIAA 2013-0287, "Stanford University Unstructured (SU²) : An open-source integrated computational environment for multi-physics simulation and design", 2013
- ¹¹F.M. White, "Viscous Fluid Flow", McGraw Hill Inc., 1974
- ¹²Antony Jameson, "Analysis and Design of Numerical Schemes for Gas Dynamics 1 Artificial Diffusion, Upwind Biasing, Limiters and Their Effect on Accuracy and Multigrid Convergence, 1995
- ¹³S. Yoon and A. Jameson, "Lower-Upper Symmetric-Gauss-Seidel method for the Euler and Navier-Stokes equations, AIAA Journal, 26(9), 1988
- ¹⁴Seulgi Yi, Hyungil Kwon, Seongim Choi, B.M. Park, Y.S. Kang, "Multidisciplinary Design Optimization of Efficient Electric Aerial Vehicle Propeller", KASA Spring Conference, 2012, Vol. 2, pp. 127~134.
- ¹⁵A. G. Journel and M. E. Rossi, "When Do We Need a Trend Model", Technical Report No.115, 1988.
- ¹⁶Dale Zimmerman, Claire Pavlik, Amy Ruggles and Marc P. Armstrong, "An Experimental Comparison of Ordinary and Universal Kriging and Inverse Distance Weighting", Mathematical Geology, Vol. 31, No. 4, 1999.
- ¹⁷Iman. R. L., Davenport. J.M., Zeigler. D. K., "Latin Hypercube Sampling (program user's guide)", 1980.
- ¹⁸Ira H. Abbott, Albert E. von Doenhoff, Theory of Wing Sections including a Summary of Airfoil Data, Dover Publications, New York, 1959.
- ¹⁹Arvel E. Gentry, "the Aerodynamic of Sail Interaction", Proceedings for the Third AIAA Symposium on the Aero/Hydrodynamics of Sailing, 1971.
- ²⁰Vincent G.Chapin, "Analysis, Design and Optimization of Navier-Stokes Flows around Interacting Sails", MDY06 International Symposium on Yacht Design and Production, 2006.
- ²¹Sangho Kim, Juan J. Alonso and Antony Jameson, "Design Optimization of High-Lift Configurations Using a Viscous Continuous Adjoint Method", 40th AIAA Aerospace Science Meeting and Exhibit, AIAA 2002-0844, 2002.
- ²²Seiki Onishi, Tsutomu Momoki and Yoshiho Ikeda, "A study on a vessel with multiple flat and hard sails to keep service speed in high winds", The 6th International Workshop on Ship Hydrodynamics IWSH2010, 2010
- ²³Anderson, John D, Introduction to Flight, McGraw-Hill. ISBN 0-07-282569-3, 2004.
- ²⁴Clancy, L. J., Aerodynamics, Pitman Publishing Limited, London ISBN 0-273-01120-0, 1975.
- ²⁵Jasbir S. Arora, Introduction to Optimum Design, ELSEVIER, pp.22, 1959.
- ²⁶B.J. Lee, H.B. Kim, M.S. Kim, Aerodynamics, Seoul, 2006, pp. 252~253.
- ²⁷Douvi C. Eleni, Tsavalos I. Athanasios, Margaritis P. Dionissios, "Evaluation of the Turbulence Models for the Simulation of the flow over a National Advisory Committee for Aeronautics NACA0012 Airfoil", Journal of Mechanical Engineering Research, 2012, Vol. 4(3), pp. 100~111.
- ²⁸Ouchi. K., Uzawa. K., "Concept Design of Wind Driven Vessel in the Era of Low Carbon Society", Proceedings for 12th Ocean Engineering Symposium, JFEOS & JASNAOE, 2009.
- ²⁹Kanai. A., Uzawa. K., Ouchi. K., "Performance Prediction of Large Sailing Vessel with Multiple Wing Sails by CFD, Wind Tunnel Test and EPP", Conference Proceedings, JASNAOE, 2011.
- ³⁰Ninami Y., Ninura T., Fujiwara Y., Ueno M., "Investigation into Underwater Fine Arrangement Effect on Steady Sailing Characteristics of a Sail Assisted Ship", Proceedings of the 13th International Offshore and Polar Engineering Conference, 2003.
- ³¹Matsumoto N., Inoue M., Sudo M., "Operatic Performance of a Sail Equipped Tanker in Wave and Wind", Second International Conference of Stability of Ships and Ocean Vehicles (STAB), pp. 451~464, 1981.

³²Woodward, F.A., “a Unified Approach to the Analysis and Design of Wing-Body Combinations at Subsonic and Supersonic Speed”, AIAA Paper, No. 65-55, 1968.

³³Ishihara M., Watanabe T., “Prospect of Sail-Equipped Motorship as Assessed from Experimental Ship ‘Daioh’”, Shipboard Energy Conservation Symposium the Society of Naval Architects and Marine Engineers, pp. 191~198, 1980.

³⁴Quao Li, Yasunori Nihel, Yoshiho Ikeda, “A Study on Performance of Cascade of Wing Sails for Sail-equipped Vessels Considering their Aerodynamic Interaction, The 6th Asia-Pacific Workshop on Marine Hydrodynamics-APHydro2012, 2012.